Introduction to Electronics and Arduino!

Basics of Arduino Circuits *Eric, Andy, Allen*

Introduction

—

- We are going over everything from the basics
- Learning *why* is just as important as *what*

For Students

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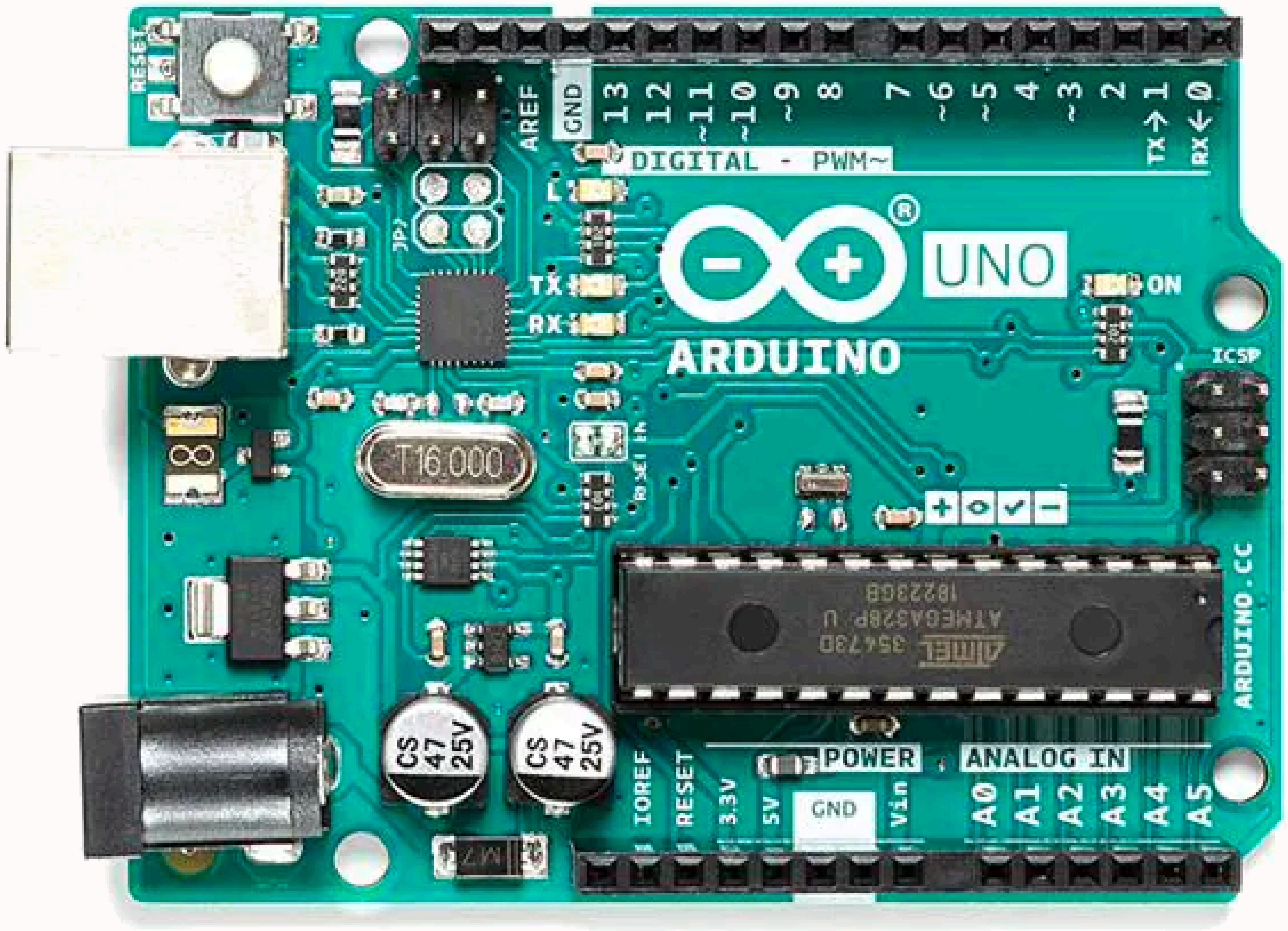
- Follow along with the workshop
- Ask questions whenever you want
- Be creative



Outline of Workshop

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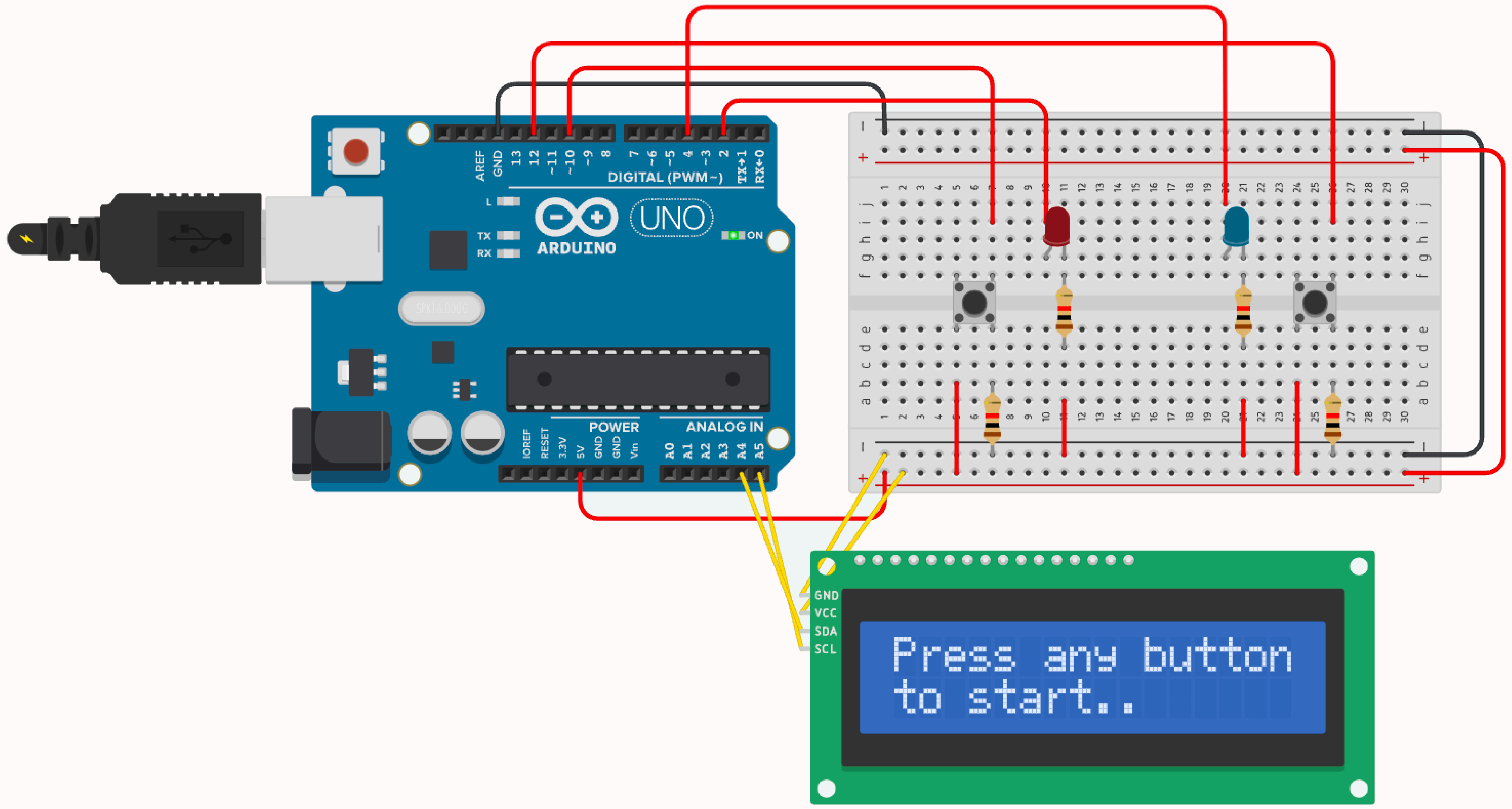
1. Electricity
2. Circuits
3. Basic Arduino Programming
4. Arduino with Circuits
5. Creating the Project



Electricity

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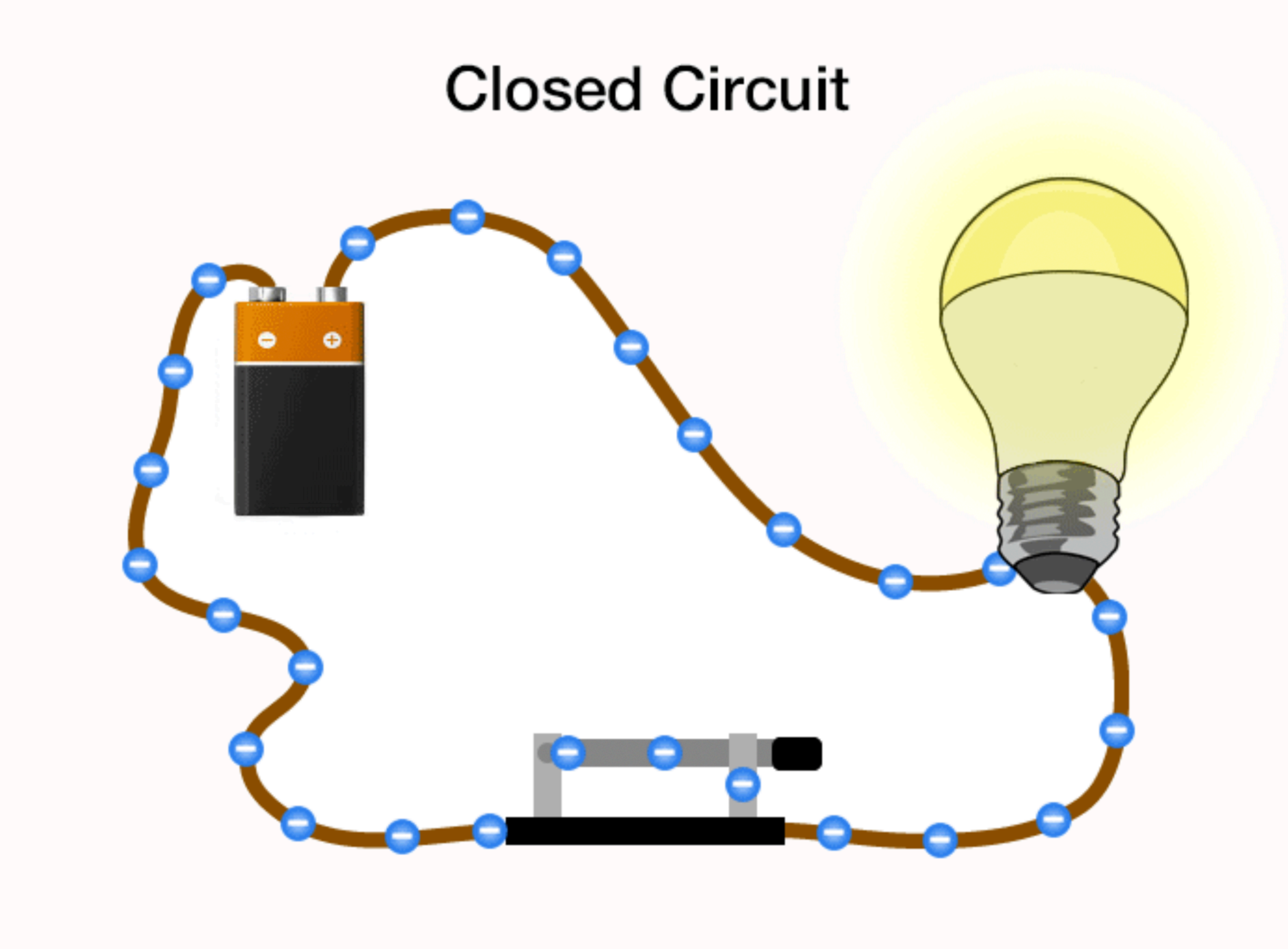
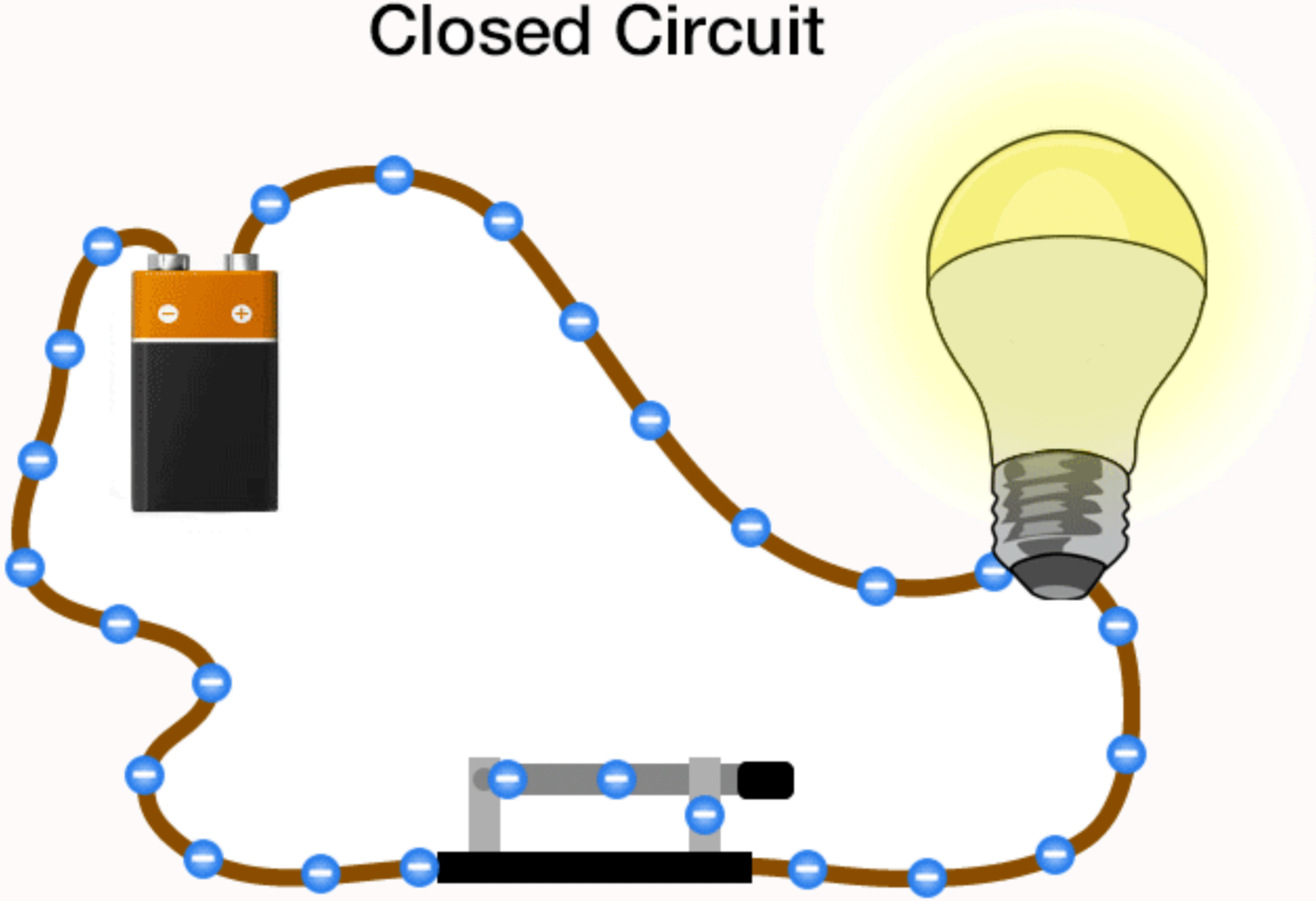
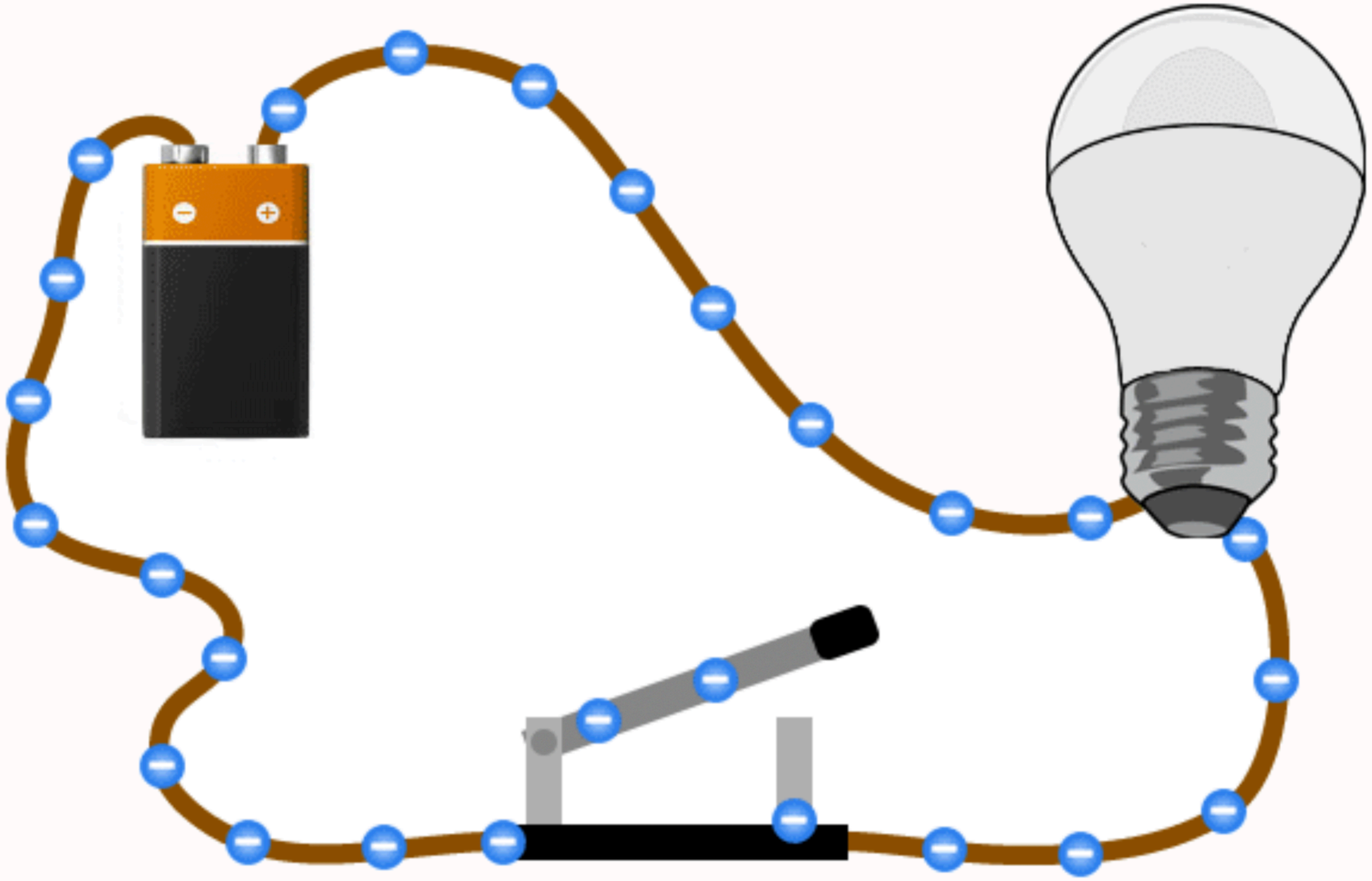
- Electrons flowing from negative to positive
- Powers components in its path (LEDs, Arduino, Light Bulb)
- Electricity can only flow if there is a path

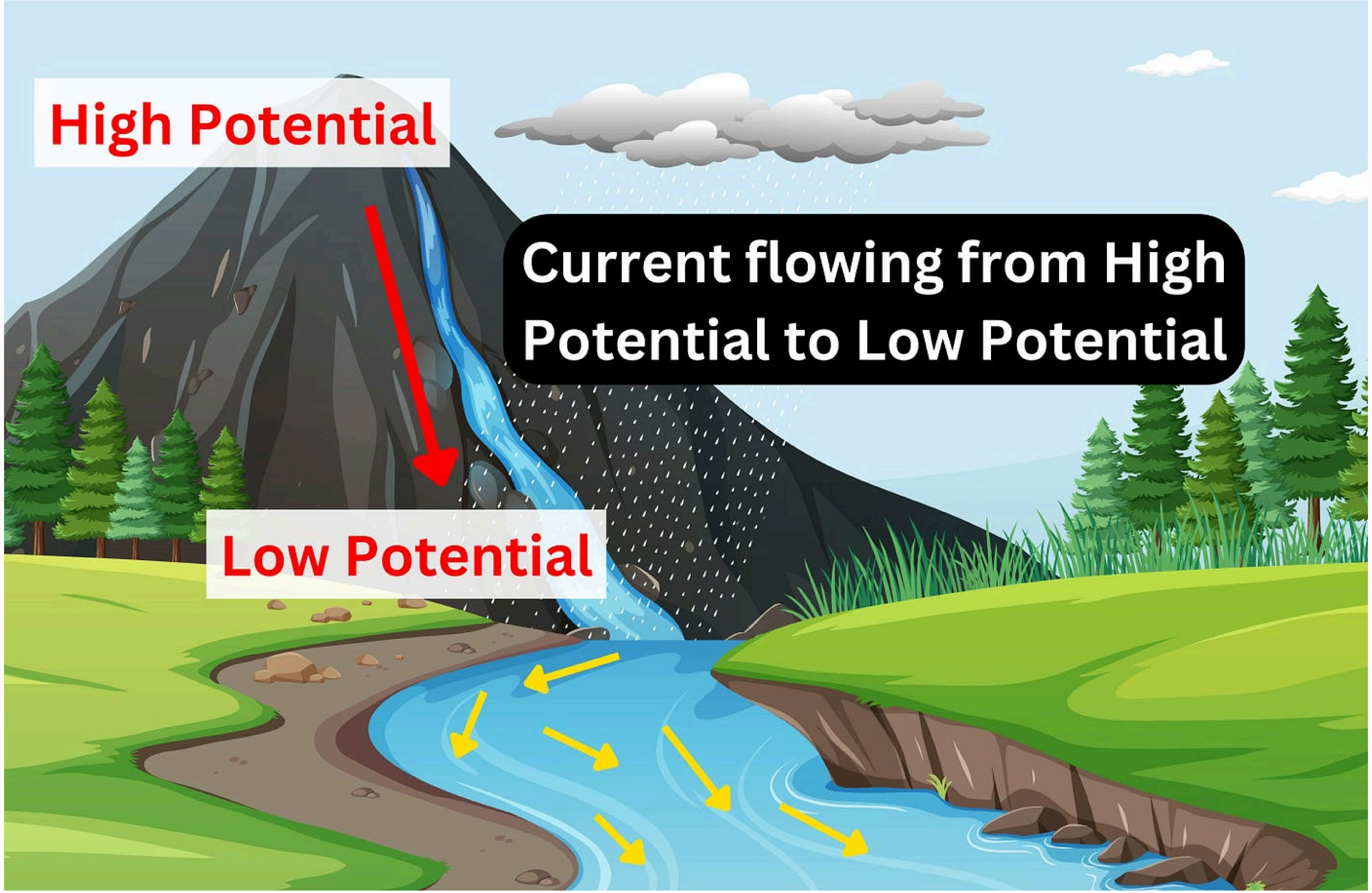


Closed Circuit



Open Circuit



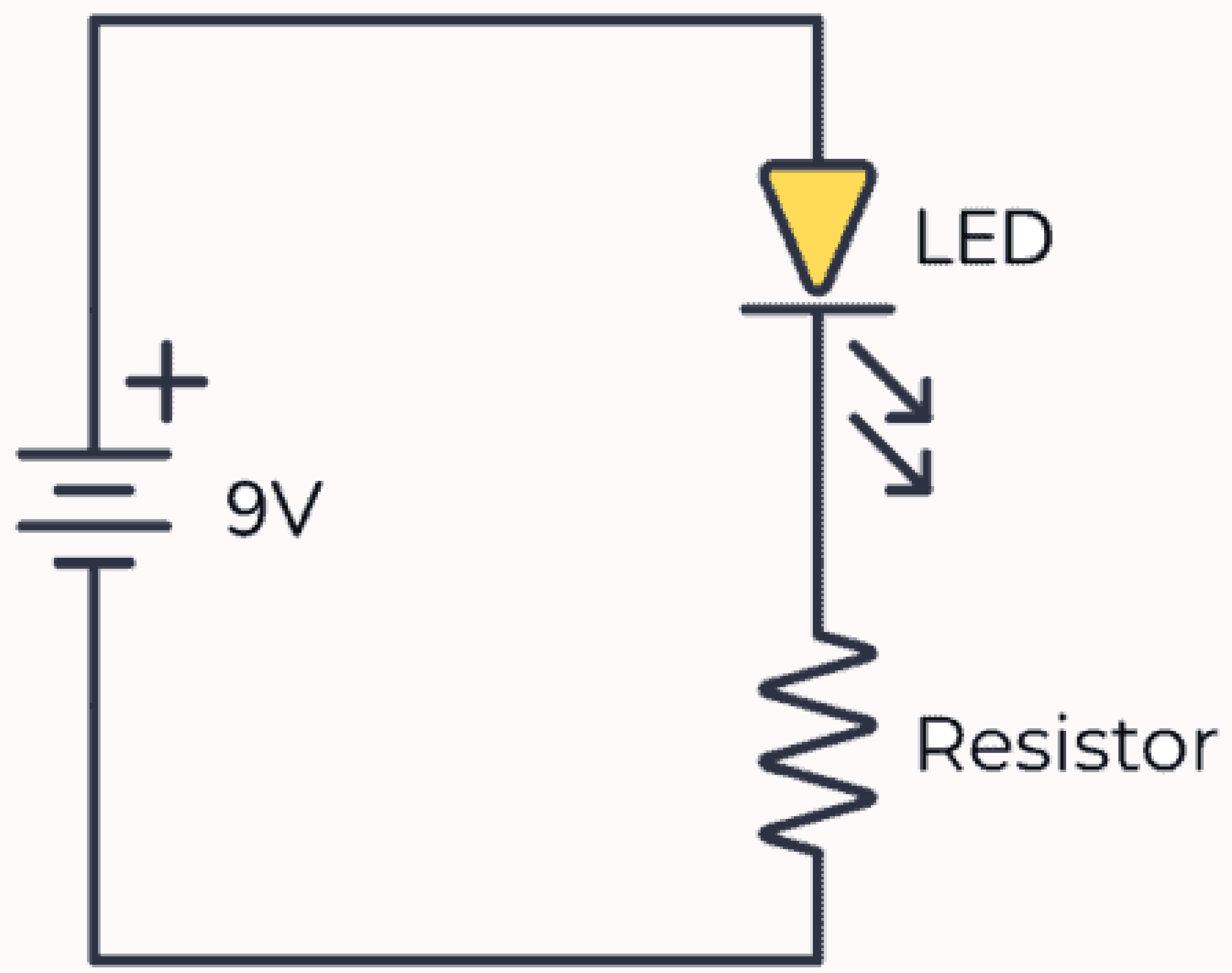


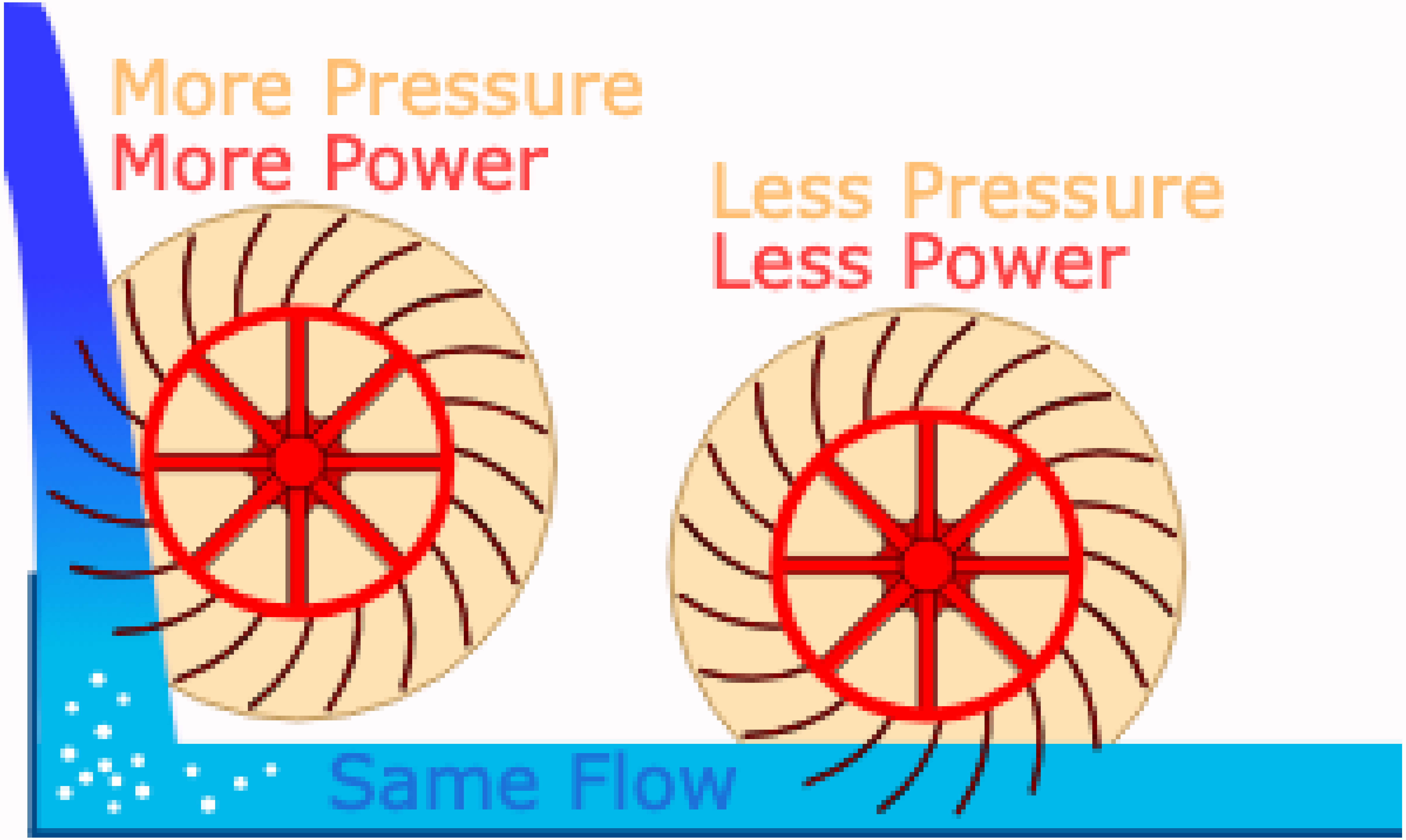
High Potential
Current flowing from High Potential to Low Potential
Low Potential

Electricity Terms

- **Current** = How much electricity flows per second
- **Voltage** = Potential electricity
- **Resistance** = How much electricity a material stops

- Ohm's Law: $V = IR \implies I = \frac{V}{R}$
- Too much current = some components overheat and break
  - LEDs, Arduino, etc.

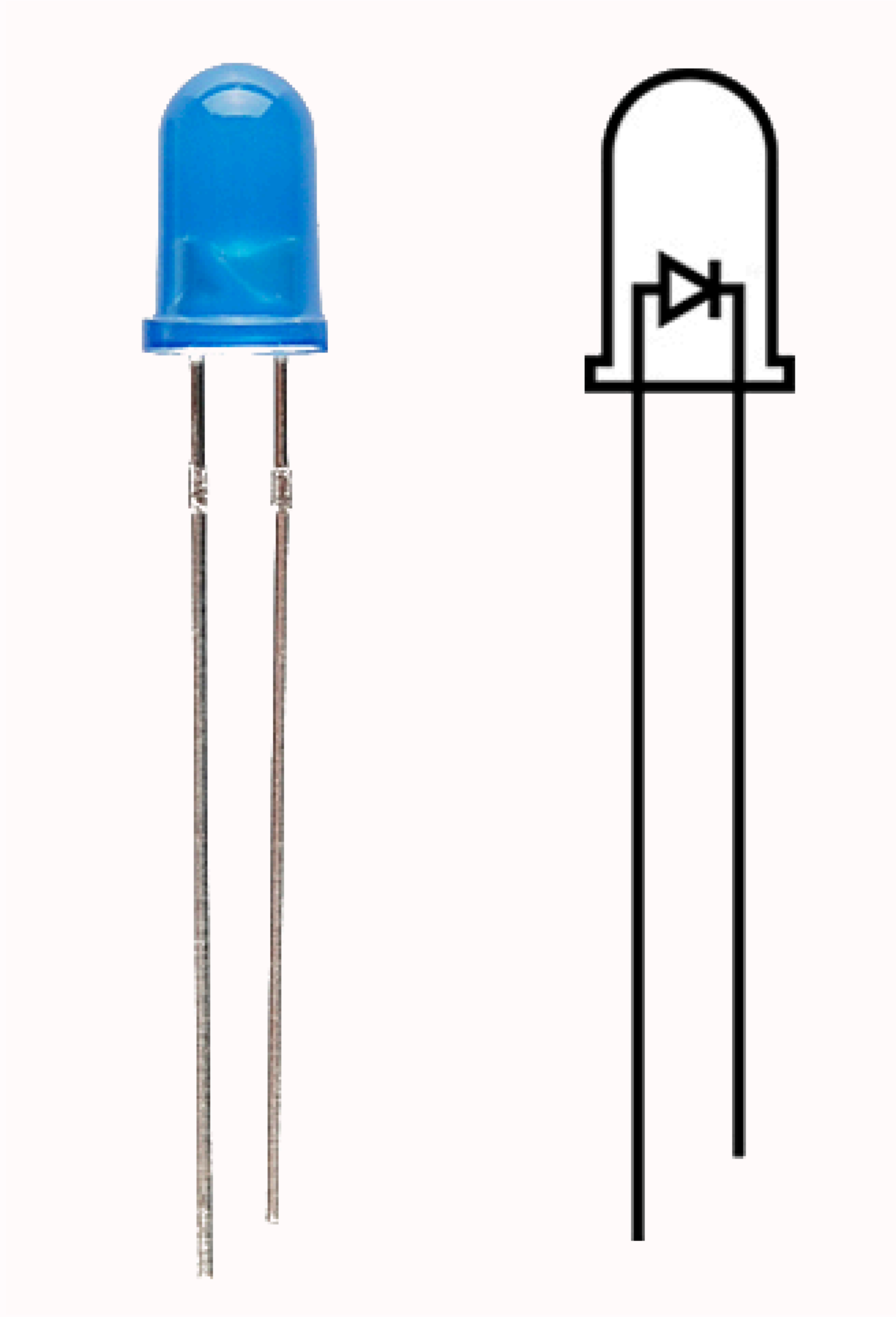



More Pressure
More Power
Less Pressure
Less Power
Same Flow

LEDs

—

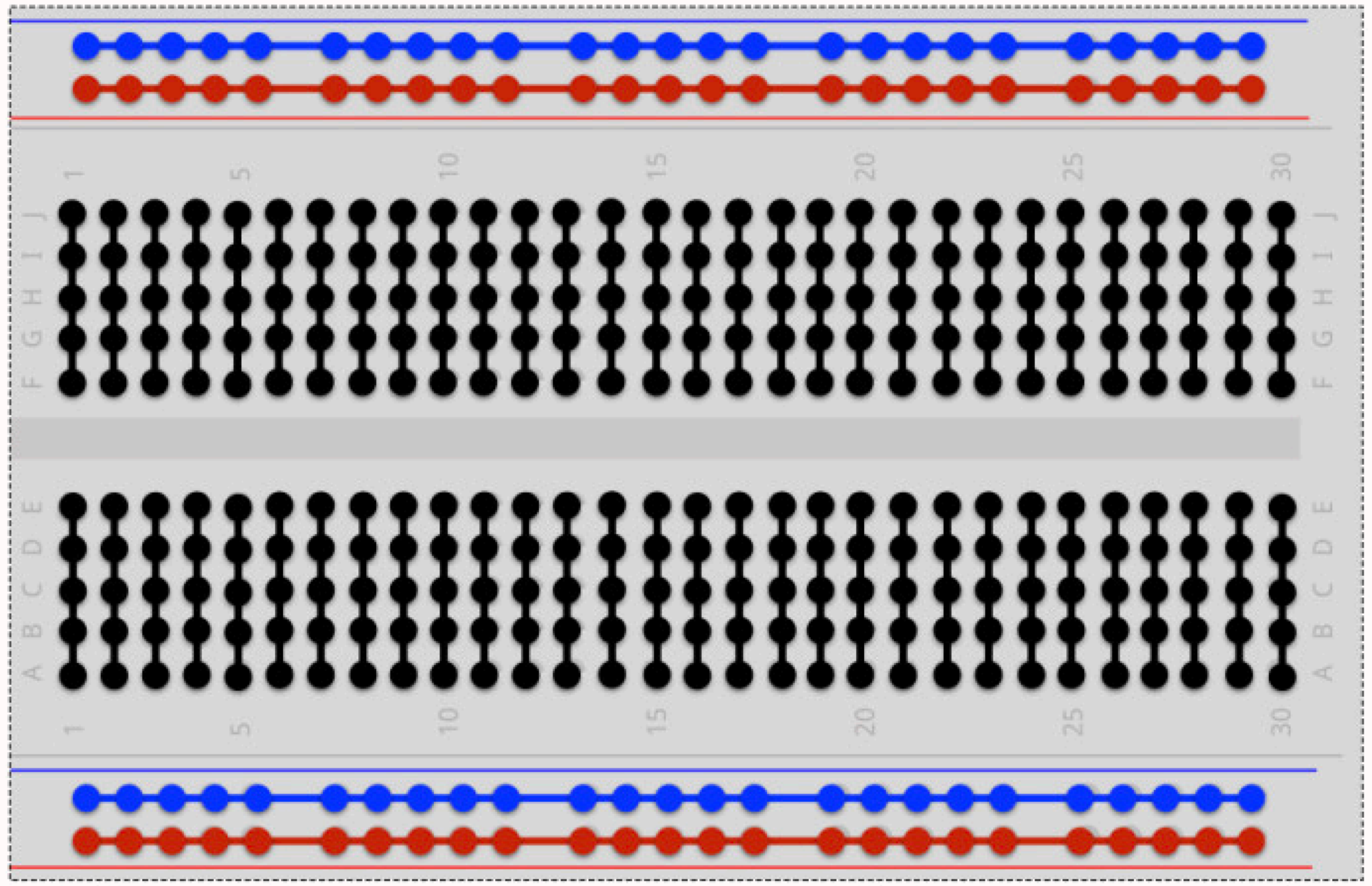
- **Light-emitting diode**
- Diode lets current go through *one direction*
- LED = *“thing that lights up when current goes through in a certain direction”*
- Longer end to high voltage (**5V**), shorter end to low voltage (**Ground**)



Circuits and Wiring

—

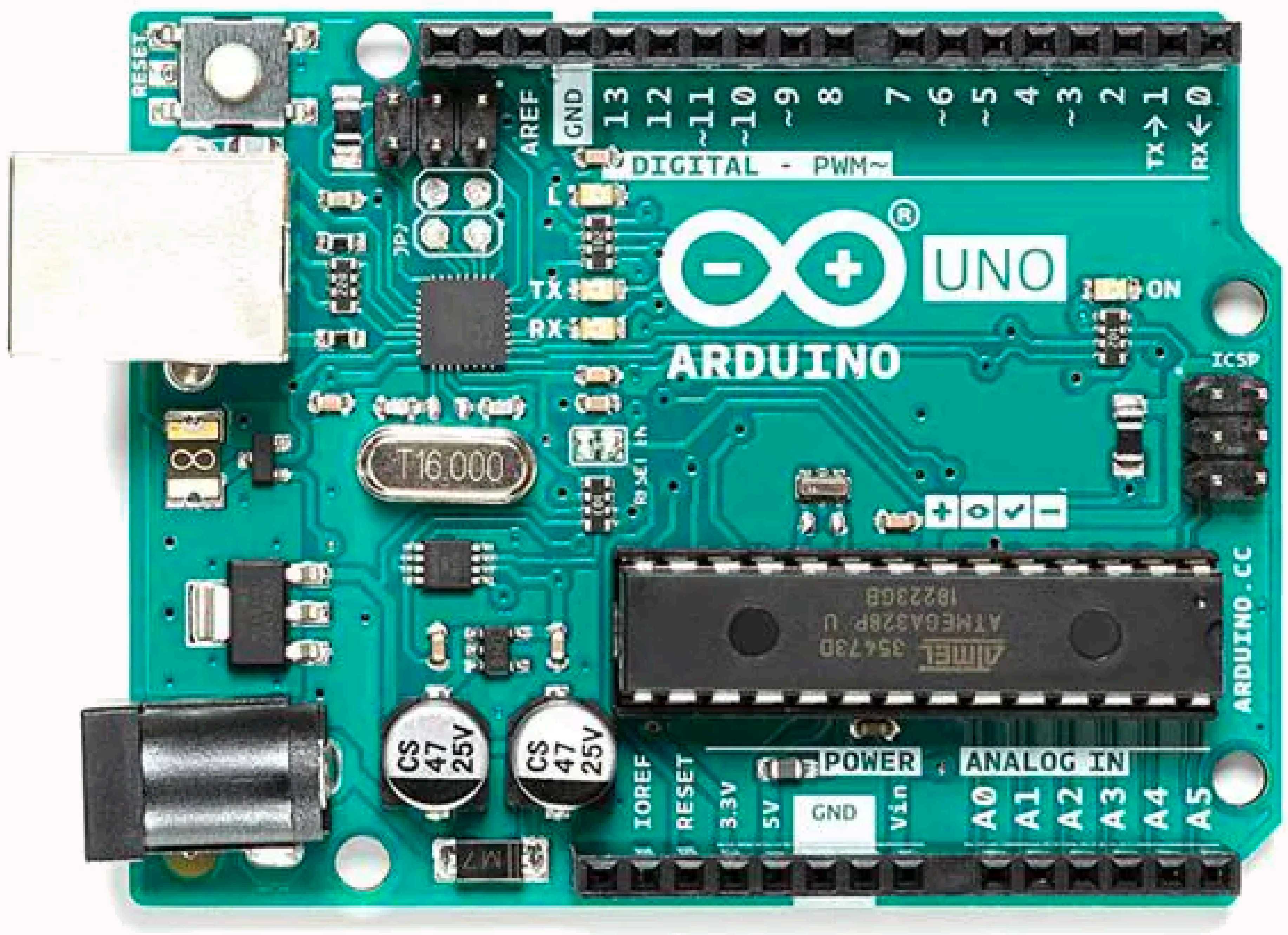
- Circuit on **breadboard**, with wires and other components
  - LEDs, Buttons, LCDs, Arduino, resistor, etc.
- Breadboards make wiring *easy to change*
  - Connects wires just by plugging in
  - Blue is connected, red is connected, black is connected
  - Great for learning & projects

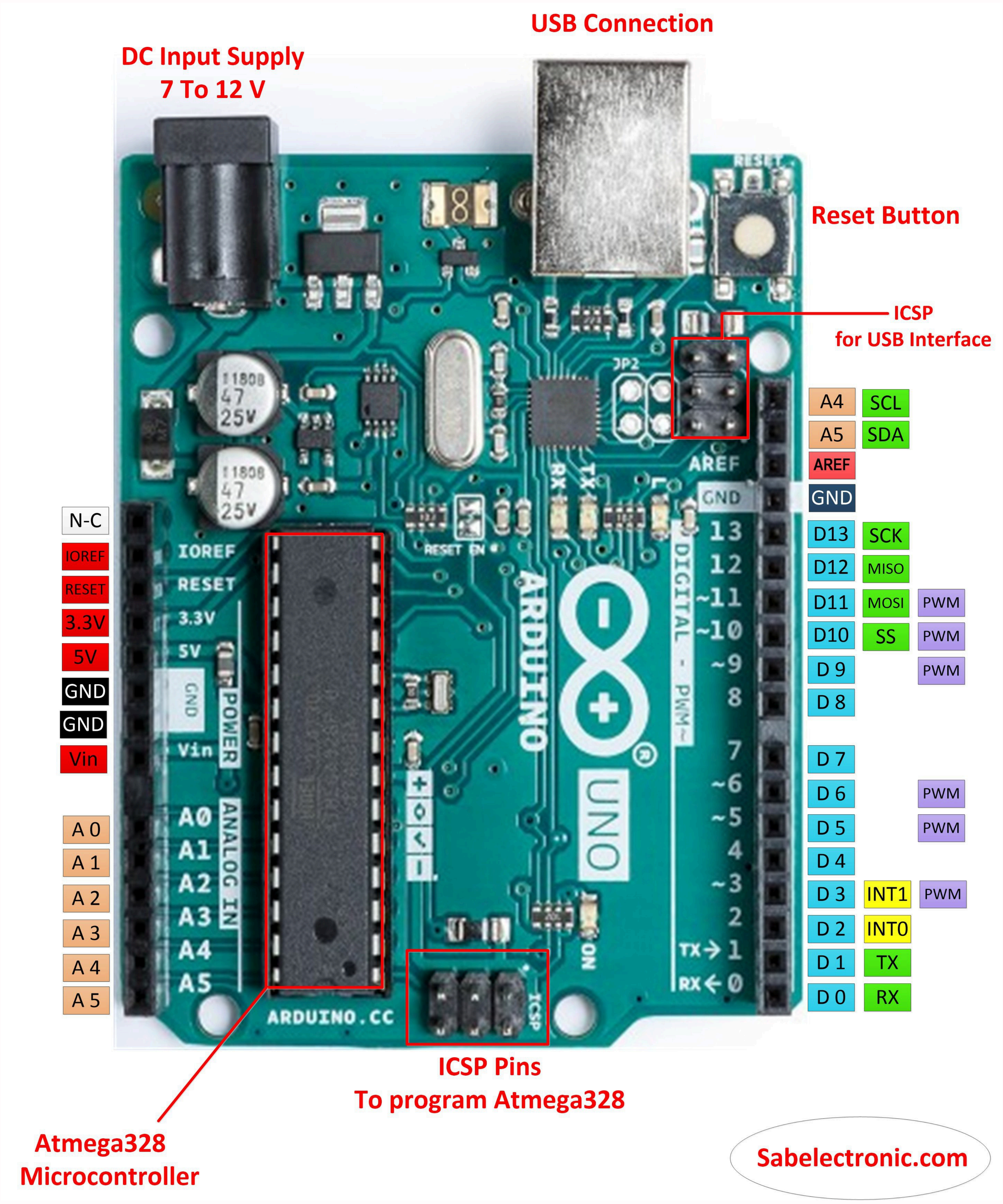


Arduino

–

- Has pins for *output* and *input*
- Has pins for **ground** and **V5** / **V3.3** (constant)
- Connects to power source (could be computer)
- Computer connects to upload code
- Arduino runs code that is uploaded, with any power source




USB Connection
DC Input Supply
7 To 12 V
Reset Button
ICSP
for USB Interface
N-C
IOREF
RESET
3.3V
5V
GND
GND
Vin
A 0
A 1
A 2
A 3
A 4
A 5
A4
SCL
A5
SDA
AREF
GND
D13
SCK
D12
MISO
D11
MOSI
PWM
D10
SS
PWM
D 9
PWM
D 8
D 7
D 6
PWM
D 5
PWM
D 4
D 3
INT1
PWM
D 2
INT0
D 1
TX
D 0
RX
ICSP Pins
To program Atmega328
Atmega328
Microcontroller
Sabelectronic.com

Arduino Code

—

- Uses Arduino language (C++ with special built-in functions)
  - *digitalWrite*(…), *delay*(…), *analogRead*(…), etc.
- Runs **setup**, then runs **loop** function until it is off

Code that turns LED on and off

```
// The setup() function runs once arduino gets power
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop() function runs over and over again
forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);
  delay(200);
  digitalWrite(LED_BUILTIN, LOW);
  delay(200);
}
```

Buttons

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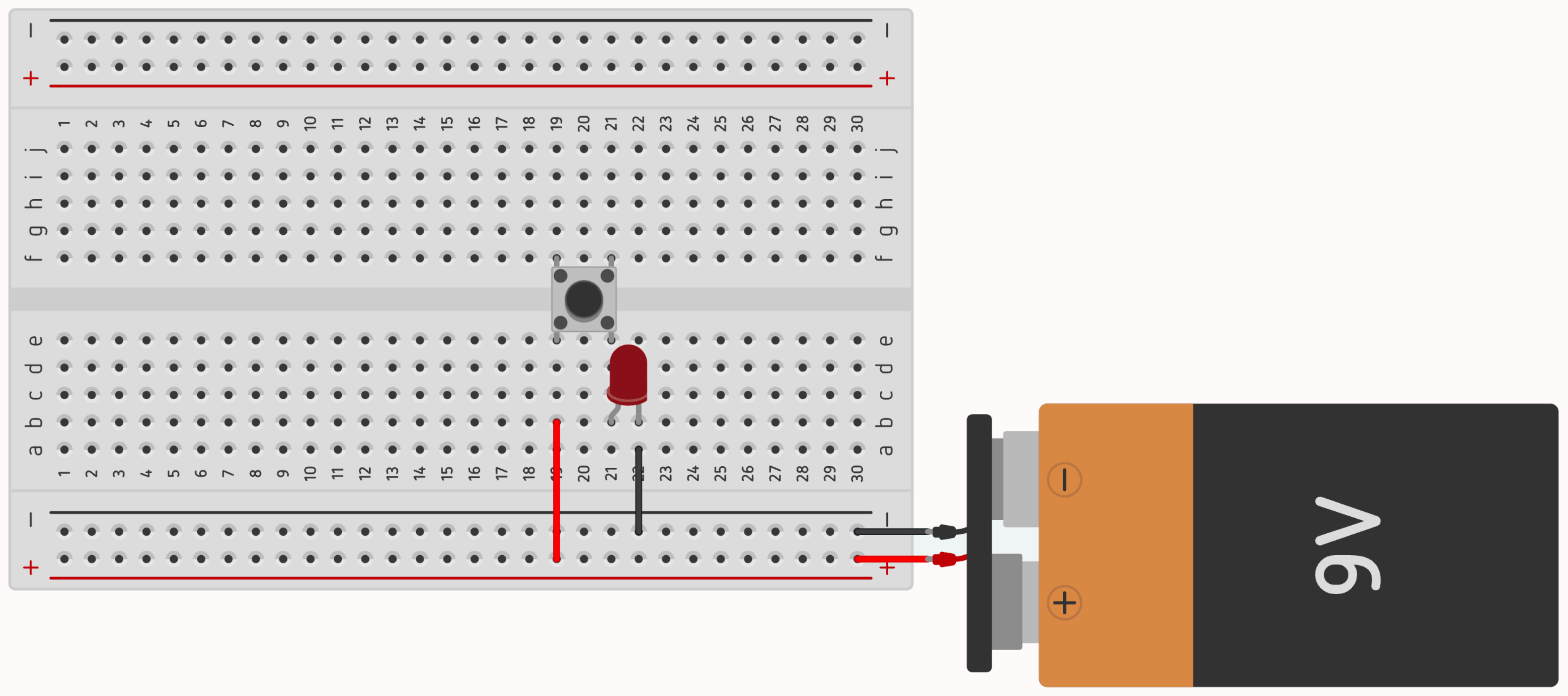
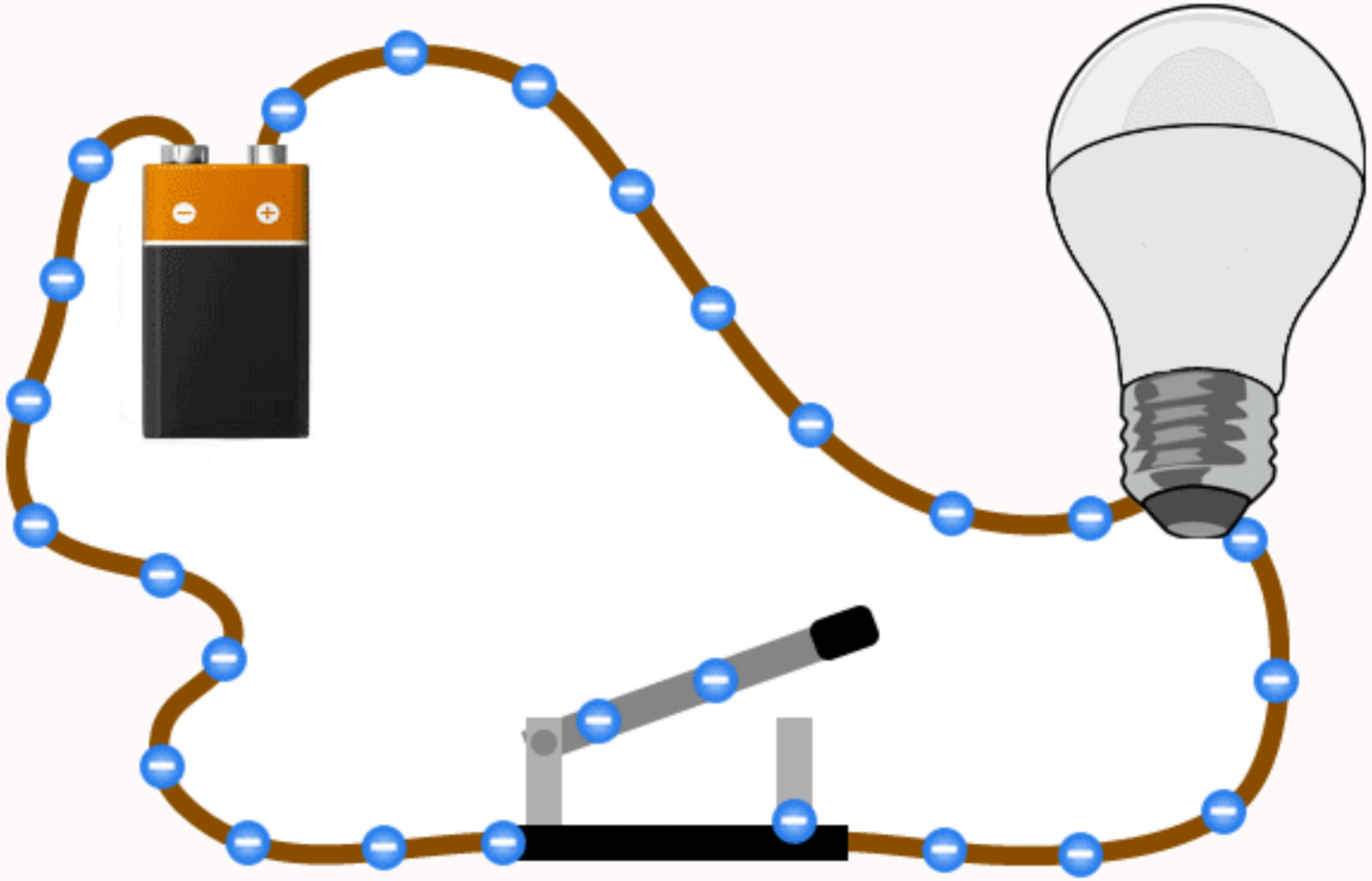
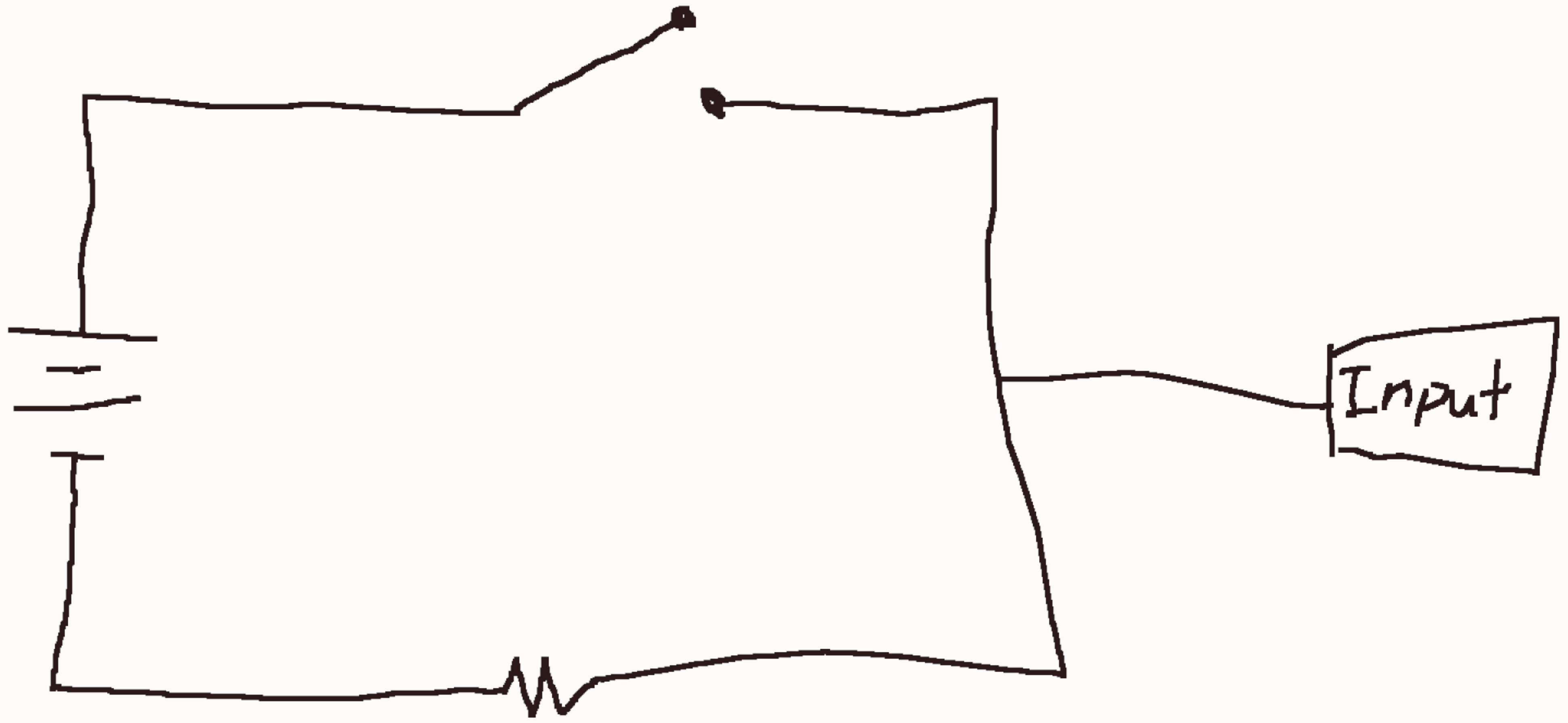
- Buttons control when circuit is open or closed
- Pushing button connects circuit
- *"Pull-Down Resistors"* are necessary when using buttons for input
  - Get rid of excess charge (sends it to **ground**)

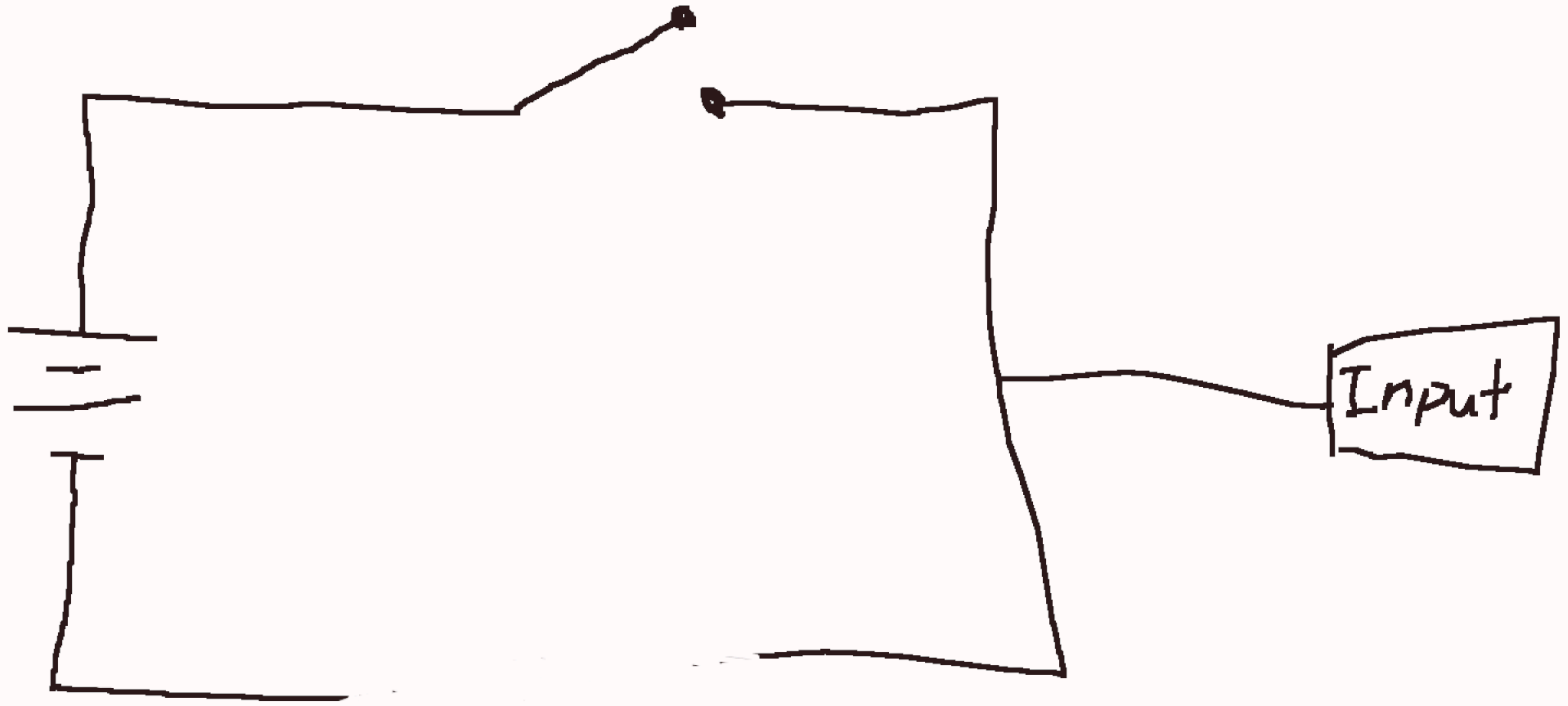


Open Circuit





(with pulldown)



(without pulldown)

(the slideshow budget ran out)

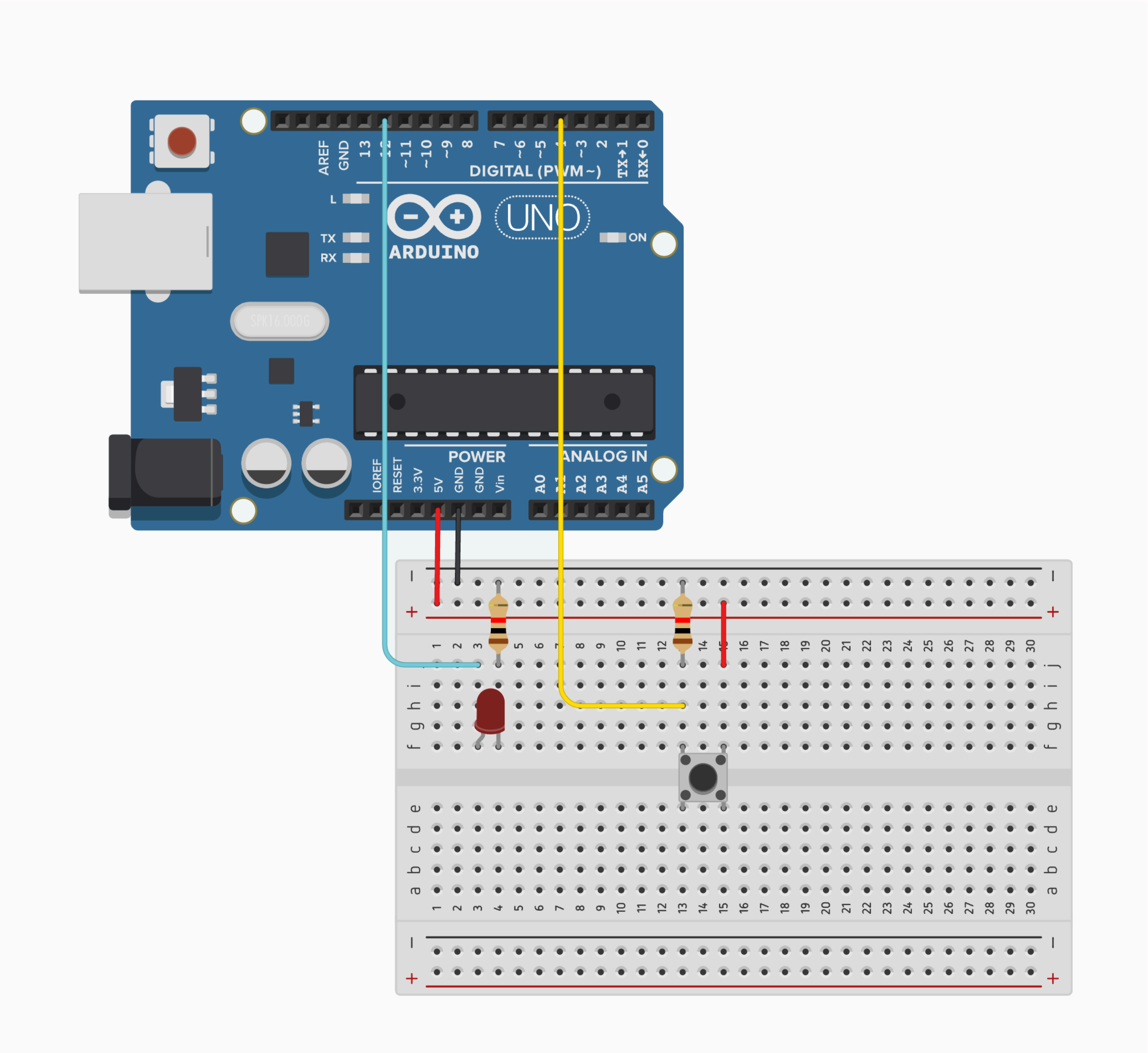
Code with Button Logic

```
const int BUTTON_PIN = 10;
const int LED_PIN = 6;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT); // set pin to input
}

void loop() {
   // runs if there is high input to the pin
  if (digitalRead(BUTTON_PIN) == HIGH) {
     // flash light once
    digitalWrite(LED_PIN, HIGH);
    delay(100);
    digitalWrite(LED_PIN, LOW);
    delay(100);
  }
}
```

Controlling LEDs

—

- Connect output to LED
- Connect input to button
- Use programming to check input
- Send voltage to LED output



```
const int LED_PIN = 12;
const int BUTTON_PIN = 4;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT);
}

void loop() {
  if (digitalRead(BUTTON_PIN) == HIGH) {
    digitalWrite(LED_PIN, HIGH);
    delay(50);
    digitalWrite(LED_PIN, LOW);
    delay(50);
  } else {
    digitalWrite(LED_PIN, HIGH);
    delay(200);
    digitalWrite(LED_PIN, LOW);
    delay(200);
  }
}
```

Can you try with 2 LEDs and 2 Buttons?

```
const int LED_A_PIN = 12;
const int LED_B_PIN = 13;
const int BUTTON_A_PIN = 4;
const int BUTTON_B_PIN = 5;

void setup() {
  pinMode(LED_A_PIN, OUTPUT);
  pinMode(LED_B_PIN, OUTPUT);
  pinMode(BUTTON_A_PIN, INPUT);
  pinMode(BUTTON_B_PIN, INPUT);
}

void loop() {
  bool a = digitalRead(BUTTON_A_PIN) == HIGH;
  bool b = digitalRead(BUTTON_B_PIN) == HIGH;

  if (a && b) {
    digitalWrite(LED_A_PIN, HIGH);
    digitalWrite(LED_B_PIN, LOW);
    delay(50);
    digitalWrite(LED_A_PIN, LOW);
    digitalWrite(LED_B_PIN, HIGH);
    delay(50);
```

```
  } else if (a) {
    digitalWrite(LED_A_PIN, HIGH);
    delay(50);
    digitalWrite(LED_A_PIN, LOW);
    delay(50);
  } else if (b) {
    digitalWrite(LED_B_PIN, HIGH);
    delay(50);
    digitalWrite(LED_B_PIN, LOW);
    delay(50);
  }

  digitalWrite(LED_A_PIN, LOW);
  digitalWrite(LED_B_PIN, LOW);
}
```

THANK YOU FOR COMING TO THE WORKSHOP

consider

learning more about programming

learning more about electronic circuits

learning more about your interests

learning more